

## Electrical · Electronic Circuit

~~Thevenin's Theorem may be stated below:-~~

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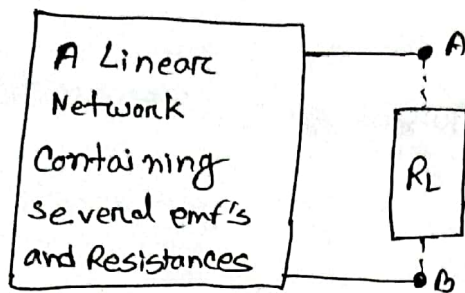
Thevenin's Theorem may be stated below:-

Any Linear Electric

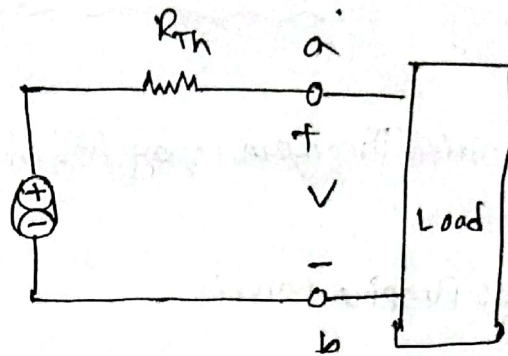
Network or complex circuit with current and voltage sources can be replaced by an equivalent circuit containing of a single independent voltage source  $V_{TH}$  and a series resistance  $R_{TH}$ .

Or

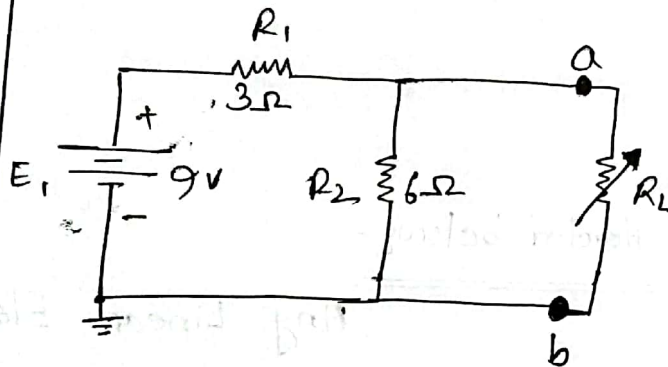
Thevenin's Theorem states that "Any linear circuit containing several voltages and resistances can be replaced by just one single voltage in series with a single resistance connected across the load".



$V_{Th}$

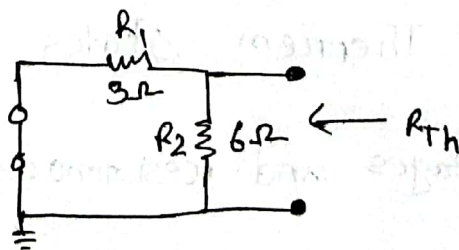
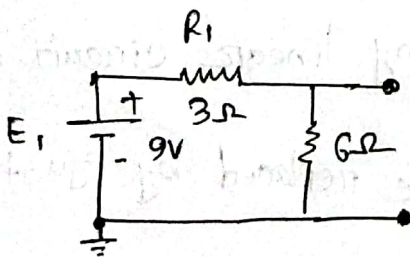


For example:



Measuring  $R_{Th}$ :

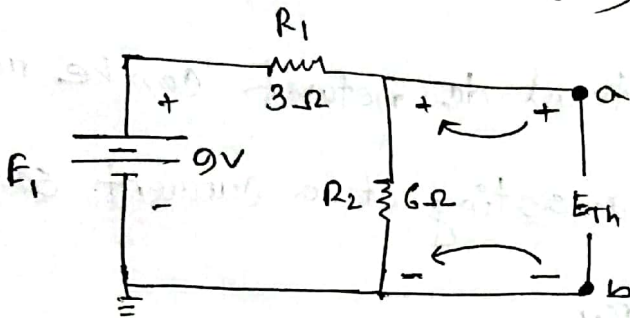
1. Open the load resistor
2. Open Current Sources and Short Voltage Sources.
3. Calculate/measure the Open Circuit Resistance. This is the Thevenin Resistance ( $R_{Th}$ )



$$R_{Th} = R_1 \parallel R_2 = \frac{3 \times 6}{3 + 6} = 2 \Omega$$

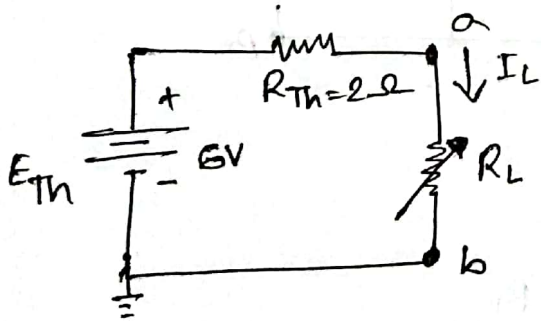
## Measuring $E_{Th}$ :

1. Open the load resistor.
2. Redraw the circuit with voltage source and current source.
3. Calculate/measure the open circuit voltage. This is the Thevenin Voltage ( $V_{Th}$ )



$$E_{Th} = \frac{R_2 E_1}{R_2 + R_1} = \frac{6 \times 9}{6 + 3} = \frac{54}{9} = 6V \quad [\text{VDR Formula}]$$

## Thevenin equivalent circuit:-



$$I_L = \frac{E_{Th}}{R_{Th} + R_L}$$

$$R_L = 2\Omega; \quad I_L = \frac{6}{2 + 2} = 1.5A$$

$$R_L = 10\Omega; \quad I_L = \frac{6}{2 + 10} = 0.5A$$

$$R_L = 100\Omega; \quad I_L = \frac{6}{2 + 100} = 0.06A$$

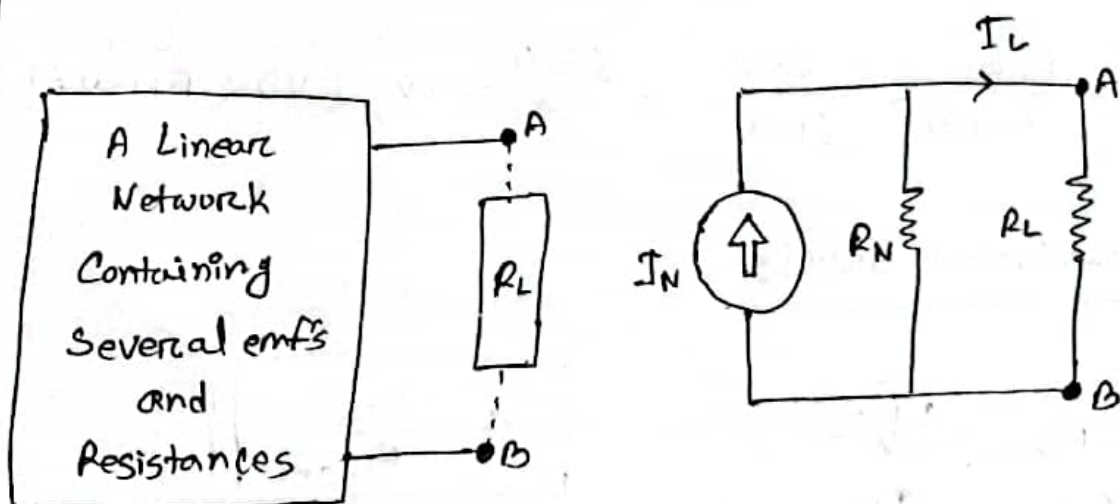
Norton's Theorem may be stated Under:-

Any Linear Electric

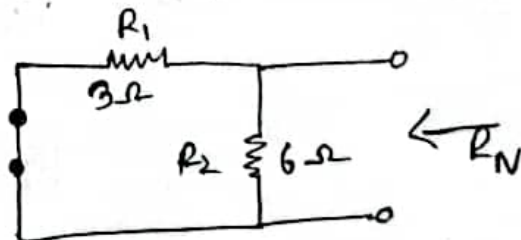
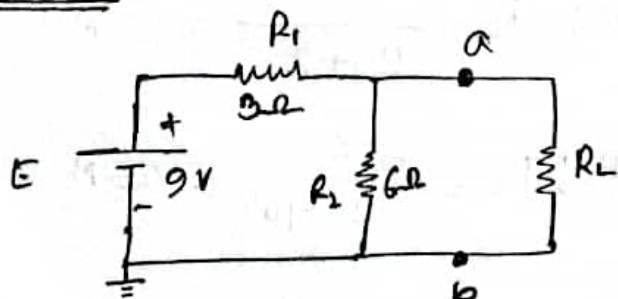
Network or complex with Current and Voltage sources can be replaced by an equivalent circuit containing of a single independent Current Source  $I_N$  and Parallel Resistance  $R_N$ .

Or

Any two-terminal linear bilateral dc network can be replaced by an equivalent circuit consisting of a current source and a parallel resistor.

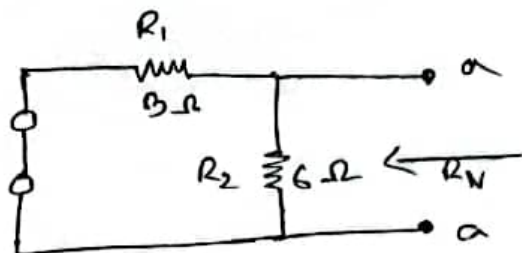


Example:



### Measuring $R_{TH}$ :

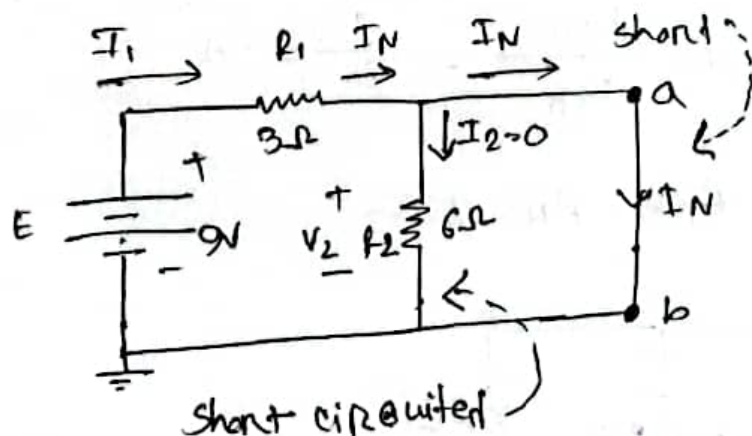
1. Open the load resistor.
2. Open current sources and short voltage sources.
3. Calculate/~~me~~ measure the open circuit Resistance. This is the Norton resistance.



$$R_N = R_1 \parallel R_2$$
$$= \left( \frac{1}{3} + \frac{1}{6} \right)^{-1} = 2 \Omega$$

### Measuring $I_N$ :

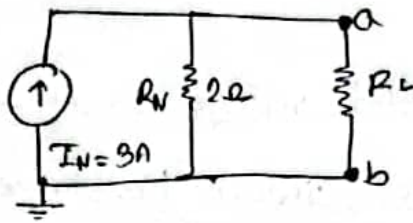
4. Short the load resistor.
5. Redraw the circuit with voltage source and current source.
6. Calculate/measure the short circuit current. This is the Norton current ( $I_N$ )



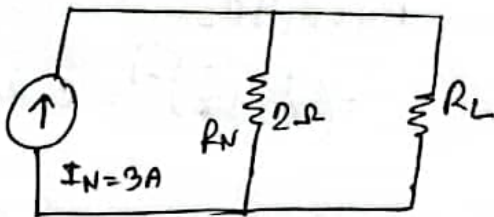
$$V_2 = I_2 R_2 = 0 \times 6 = 0V$$

$$I_N = \frac{E}{R_1} = \frac{9}{3} = 3A$$

Norton equivalent circuit:-

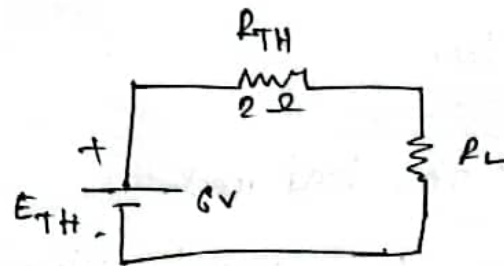


Converting norton to thevenin:

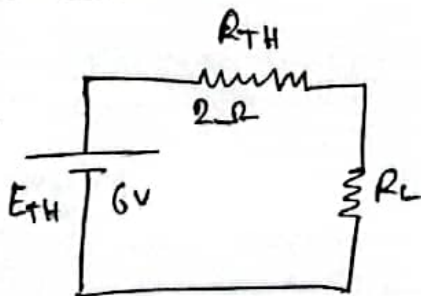


$$E_{TH} = I_N R_N = 3 \times 2 = 6V$$

$$R_N = R_{TH} = 2\Omega$$

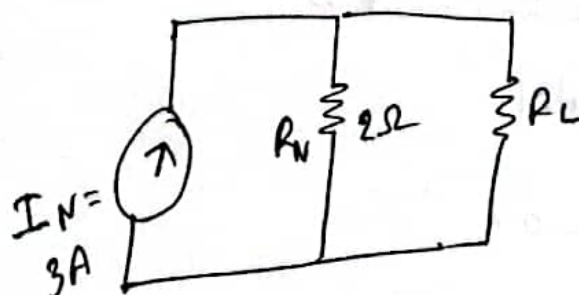


Converting thevenin to norton:



$$I_N = \frac{E_{TH}}{R_{TH}} = \frac{6}{2} = 3A$$

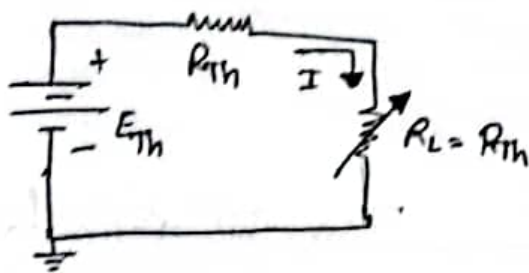
$$R_{TH} = R_N = 2\Omega$$



Q: State and explain maximum power transfer theorem with proper circuit diagram? Show that for maximum power the efficiency is 50% with efficiency curve?

Maximum power transfer statement: A load will receive maximum power from a network when its resistance is exactly equal to the Thevenin resistance of the network applied to the load, That is

$$R_L = R_{TH}$$



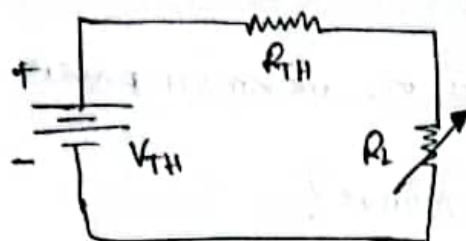
$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{E_{Th}}{R_{Th} + R_{Th}} = \frac{E_{Th}}{2R_{Th}}$$

Then substitute into the power equation:

$$\begin{aligned} P_L &= I_L^2 R_L = \left( \frac{E_{Th}}{2R_{Th}} \right)^2 (R_{Th}) \\ &= \frac{E_{Th}^2 R_{Th}}{4R_{Th}^2} \end{aligned}$$

and,

$$P_{Lmax} = \frac{E_{Th}^2}{4R_{Th}}$$



From the above circuit, the current flowing through the load, "I" is given as

$$I = \frac{V_{TH}}{R_L + R_{TH}}$$

Power transferred to the load,

$$P_L = I^2 R_L = \left( \frac{V_{TH}}{R_L + R_{TH}} \right)^2 \cdot R_L$$

Use this formula,

$$\frac{d}{dx} \left( \frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad [\text{use this formula}]$$

$$\frac{\partial P_L}{\partial R_L} = V_{TH}^2 \left[ \frac{(R_L + R_{TH})^2 - 2R_L(R_{TH} + R_L)}{(R_L + R_{TH})^4} \right] = 0$$

$$\Rightarrow (R_L + R_{TH})^2 = 2R_L(R_{TH} + R_L)$$

$$\Rightarrow (R_L + R_{TH})^2 = 2R_L(R_{TH} + R_L)$$

$$\Rightarrow R_L + R_{TH} = 2R_L$$

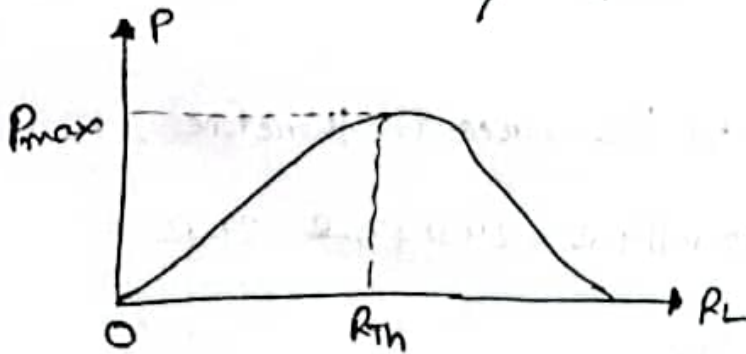
$$\Rightarrow R_L = R_{TH}$$

Again,  $R_L = \left( \frac{V_{Th}}{R_{Th} + R_{Th}} \right) R_{Th}$

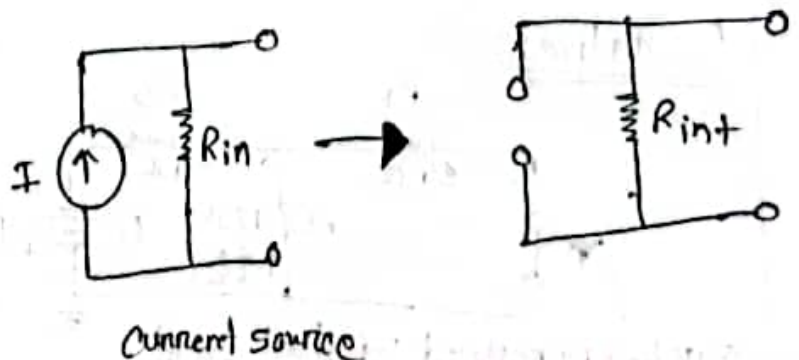
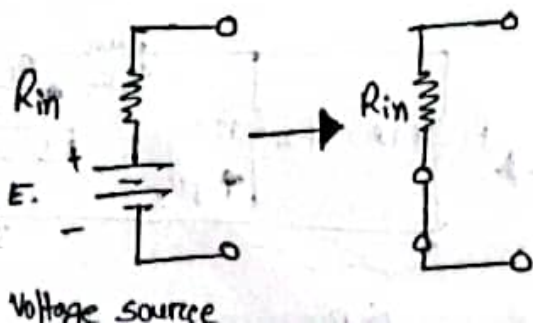
$$\therefore R_L = \frac{V_{Th}}{4R_{Th}}$$

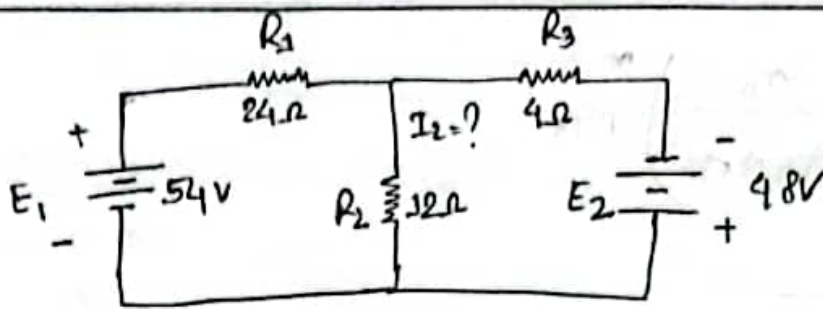
$$P_{in} = V_{Th} \times I = V_{Th} \times \frac{V_{Th}}{2R_{Th}} = \frac{V_{Th}^2}{2R_{Th}}$$

$$\therefore \eta = \frac{P_L}{P_{in}} = \frac{V_{Th}^2 / 4R_{Th}}{V_{Th}^2 / 2R_{Th}} = 50\%$$

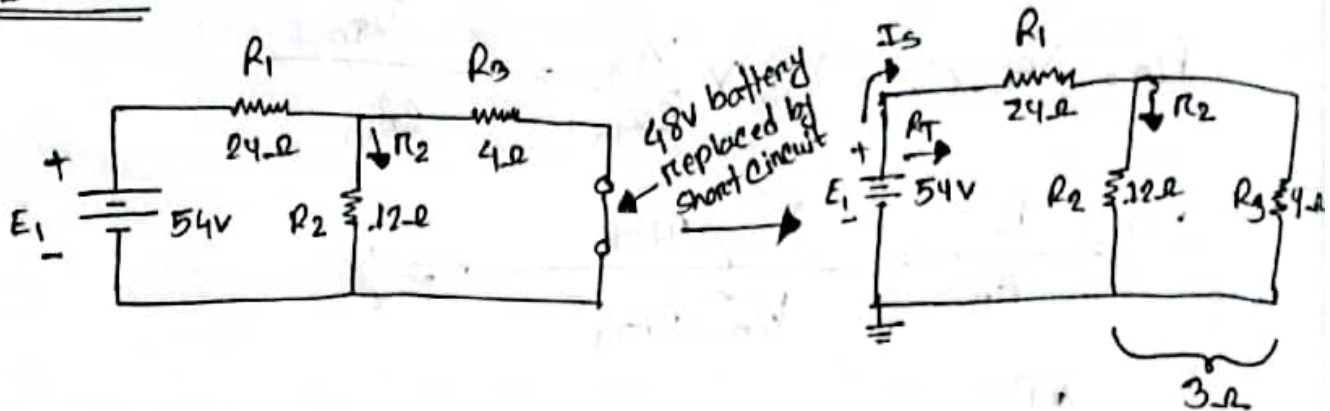


Superposition Theorem: The current through, or voltage across, an element,  $a$  in a linear bilateral network equal to the algebraic sum of the currents or voltages produced independently by each source.





$E_1$  active:



The total resistance seen by the source is therefore

$$R_T = R_1 + R_2 \parallel R_3 = 24\Omega + 12\Omega \parallel 4\Omega = 24\Omega + 3\Omega = 27\Omega$$

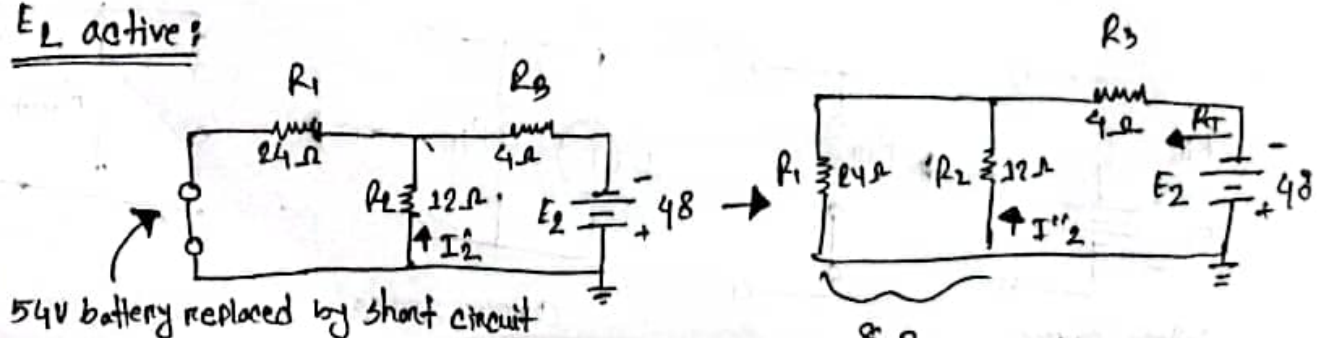
and the source current is,

$$I_S = \frac{E_1}{R_T} = \frac{54V}{27\Omega} = 2A$$

Using the current divider rule results in the contribution to  $I_2$  due to the 54V source:

$$I_2' = \frac{R_3 I_S}{R_3 + R_2} = \frac{4 \times 2}{4 + 12} = 0.5A$$

$E_2$  active:



Therefore, the total resistance seen by the 48V source is

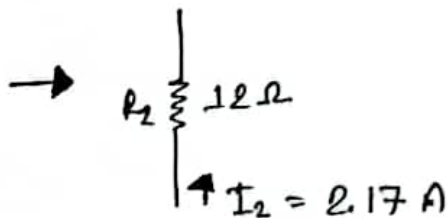
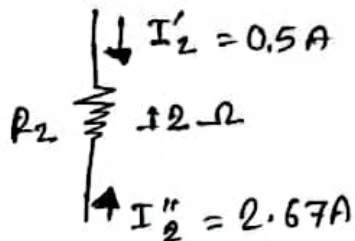
$$R_T = R_3 + R_2 \parallel R_1 = 4 + 12 \parallel 24 = 4 + 8 = 12 \Omega$$

and the source current is

$$I_S = \frac{E_S}{R_T} = \frac{48}{12} = 4 \text{ A}$$

Applying the current divider rule results in,

$$I_2'' = \frac{R_1 (I_S)}{R_1 + R_2} = \frac{24 \times 4}{24 + 12} = 2.67 \text{ A}$$



$$I_2 = I_2'' - I_2' = 2.67 - 0.5 = 2.17 \text{ A}$$